Creditors' Influence on Firm Investment Policy: An Empirical Analysis of Debt Covenant Violations

(GitHub version)

@isabellftz

Abstract

We find that debt covenant violations trigger changes in firms' investment policies. We investigate the effect of violations of accounting covenants on four investment measures, using data on non-financial U.S. firms between 1997 and 2009. Our results confirm the findings of Nini et al. (2012): debt covenant violations have a statistically and consistently negative effect on growth in total assets, net property, plant & equipment and the ratio of capital expenditures to asset ratio within one year post-violation. The effect is smaller in size for one-time violator firms. We further provide evidence that debt covenant violations do not lead to changes in firms' research and development expenditures scaled by total assets.

1 Introduction

Financial covenants, or debt covenants, legally bind the company to meet or maintain certain thresholds in accounting-based ratios or metrics to ensure firms' financial performance remains steady and loan repayments to creditors[1] can be made. Included in credit agreement contracts such as those of public debt (e.g., Myers (1977)), private debt (e.g., Roberts and Sufi (2009)), and private equity (e.g., Kaplan and Strömberg (2003)) firms, financial covenants are a ubiquitous element of corporate financial policy (Chava and Roberts (2008)). Roberts and Sufi (2009) examined 3,603 private credit agreements and found that 96.5% included at least one financial covenant. In recent years, theoretical and empirical research on financial contracting have provided ample evidence that debt covenants, or more precisely the event of a *debt covenant*

[1] The terms *creditor* and *lender* are used synonymously in this paper.

violation, are ideal for studying the influence of creditors on firms' policies (Nini et al. (2012)). Under the terms of credit agreements, a financial covenant violation constitutes a *technical* event of default. It grants creditors the right to immediately accelerate loan payments, terminate further credit facilities or to withdraw from the credit agreement (Roberts and Sufi (2009)). In practice, financial covenant violations are rarely followed by such drastic steps. Roberts (2015) found that in three out of four cases a technical default is followed by contract renegotiation (Roberts (2015)) and ultimately by waivers (Chen and Wei (1993)) of the violation. As an illustrative example, the annual report of Boston Biomedica Inc. for fiscal year 2003, required by the U.S. Securities and Exchange Commission, includes the following covenant violation:

> Interest expense, incurred primarily on the Company's outstanding mortgage on the Company's headquarters located in West Bridgewater, Massachussetts, increased [by] $43,000 in 2003 as compared to 2002. The increase was a result of a mortgage covenant waiver fee as the Company failed to meet its debt service coverage and other covenants for the year ended December 31, 2002.[2]

Renegotiation can occur not only after a covenant violation has occurred, but also ex-ante, i.e. in anticipation of a violation. In any case, creditors enter these renegotiations with an enhanced bargaining position. In our example, creditors used their acceleration right to extract amendment fees post-violation. The shift in bargaining power towards lenders, describes Ferreira et al. (2018) as "an increase in creditor control rights" (p. 2385). The question arises as to how and to what extent the strengthening of creditor control rights brought about by financial covenant violations can influence corporate financial policies. In this paper, we show that covenant violations induce immediate changes in firms' investment policies that prove to be significant even one year post-violation. These results suggest that renegotiated credit agreements have been modified to be more restrictive of firm investment decision-making, whereas creditors play an active role. We further provide evidence that the extent of creditor influence on a firm's investment policy is related to the number of debt covenant violations. In particular, we document that debt covenant violations result in significantly fewer actions undertaken by creditors against firms that classify as one-time violators. Our main findings include that the ratio of capital expenditures to total assets and growth in assets and net property, plant and equipment decline after a debt covenant violation within one year, whereas the decrease is smaller in size for firms that have reported one covenant violation throughout our study period, 1997-2008. The effect of debt

[2]Excerpt is from the 10-K SEC filing of former Boston Biomedica Inc. (since 2004 renamed Pressures Biosciences Inc.) for fiscal year ended December 31, 2003 available at https://ir.pressurebiosciences.com/annual-reports/content/0001104659-04-008715/0001104659-04-008715.pdf.

covenant violations on asset growth rate magnifies for frequent violator firms. Moreover, we find that debt covenant violations do not evoke changes in firms' ratio of research and development (R&D) expenses to total assets one year after the violation occurred, suggesting that creditors restrain from exerting their control over firms' innovation expenditures. Our paper relates to a growing body of literature that highlights the role of creditors in corporate financial policies, the closest of which examine the effect of debt covenant violations on firm outcomes. While Nini et al. (2012) shows that capital investment decreases sharply following a financial covenant violation, Roberts and Sufi (2009) finds complementary evidence that firms' debt policies follow the same trend. Nini et al. (2012) give extensive analysis, showing that debt covenant violations are associated with changes in investment, financing, operating performance policy, as well as stock price performance. Nini et al. (2012) and Ferreira et al. (2018) provide further evidence that violations trigger changes in the firm's board management composition, suggesting "that creditors also exert informal influence on corporate governance" (Nini et al. (2012), p. 1714). Our work makes several contributions to this literature. Our focus of firm investment policy allows us to review Nini et al. (2012) results. By examining post-violation changes on R&D expenses relative to total assets, we attempt to build upon Chava and Roberts (2008) study which left this measure of corporate innovation policy subject to future research. Finally, note that the previous literature has studied how and to what extent creditors exhibit influence on firms' governance in response to a debt covenant violation for all violator firms. In the present paper, however, we ask –in the context of firm's investment policy– whether creditors actions differ for one-time violator firms, and how. The remainder is structured as follows: Section 2 presents the main hypothesis; Section 3 describes the data set construction and composition; Section 4 presents our econometric strategy and the results for each hypothesis; Section 5 concludes.

2 Hypothesis Development

The presence of debt covenants enables the transfers of control rights from management of the firm to its creditors through which financing frictions, such as renegotiation costs, impact firm's investment. We ask if the change in governance triggered by the violation implies a long-lasting impact on firm's investment policy. Following Nini et al. (2012), we employ three measures that proxy firm's investment policy: Total assets, net property, plant & equipment (PPENT) and capital expenditures scaled by average assets. We expect to approve the findings by Nini et al. (2012) and Chava and Roberts (2008), which suggest a negative relationship between all three measures and the event of a debt covenant violation. In addition, we include research and development expenses (R&D) scaled by total assets into our set of measures. We expect this ratio to decline in response to a covenant violation, as “R&D expenditures are a class of discretionary investing activities that is particularly susceptible to reduction if a company suffers from a loss in value” (Smith (1993), p. 299).

We claim:

Hypothesis I: **Ceteris paribus, financial covenant violations are followed by reductions in the growth rate of total assets and PPENT, and decreases in capital expenditures-to-average assets ratio and R&D expenditures-to-assets ratio four quarter after the new financial covenant violation was reported.**

We next attempt to answer the question if the effect alters in size for firms that have violated a debt covenant violations (exactly once) more than once between 1997 and 2008. We surmise that the frequent violator firms are stronger affected by the control rights transfer accompanying covenant violations, while one-time violator firms are weaker affected. We assume that reformed corporate boards and the resulting implementation of creditor-friendly policies (Ferreira et al. (2018)) lead to a “policy of acridity” for frequent-violators and of “mercy” for one-time violator. We claim:

Hypothesis II: **Ceteris paribus, the effect of financial covenant violations on all four firms’ investment measures is larger for firms that have repeatedly reported covenant violations (frequent violators) and smaller for firms that reported only one covenant violation (one-time violators) between the second quarter of 1997 and the second quarter of 2008.**

Testing these two hypothesis empirically is the aim of our paper.

3 Data

The empirical analysis uses debt covenant violation data provided by Nini et al. (2012)[3] and firms’ accounting data from S&P Capital IQ database and Compustat[4] database.

[3]The data was made available as supplementary material to Nini et al. (2012) at `https://amirsufi.net/chronology.html`. File name of Stata data set and direct download: cstatviolations_nss_20090701.dta.

[4]I thank my Professor for providing me with a Compustat datafile that included five variables, which were missing on S&P Capital IQ database, but were indispensable for my analysis: DLCQ, TXDITCQ and PRCCQ and the two merging key variables, calendar and fiscal year-quarters

3.1 Sample Construction

To construct our sample, we start with the universe of all non-financial U.S. firms on S&P Capital IQ during the fiscal[5] years 1996 to 2009, which the Compustat additional firms' data is merged onto. Three sample restrictions are applied. First, the data set is limited to the sample period starting at the second quarter of 1996 and ending at the second quarter of 2009. Second, the sample is filtered for all firms with an average total assets greater than $10 million in the year 2000. Third, all firm-quarter observations with missing values for total assets, total sales, common shares outstanding, closing share price, and the calendar year-quarter are dropped. Variable definitions are provided in the Appendix: List of Abbreviations. All sample restrictions are applied according to Nini et al. (2012) in order to replicate the data set used in Nini et al. (2012) as precisely as possible. This leaves a sample of 8,166 non-financial U.S. firms with 230,701 quarterly observations on firm's accounting data from the second (first fiscal) quarter of 1996 to the second (forth fiscal) quarter of 2009.

Before merging the firm-quarter data set of financial covenant violations, provided by Nini et al. (2012) onto this sample, two filters are employed on the former: First, all firms are assigned to three groups: non-violators, one-time violators and frequent violators. *One-time violator* (*Frequent violator*) is a dummy variable that takes a value of one if a given firm violates one (more than one) debt covenant in any given quarter from the second quarter of 1997[6] to the end of the sample period of the financial covenant violations data set, the second quarter of 2008. Given a firm is a non-violator, the dummy variables *Frequent violator* and *One-time violator* both take zero as their values. The definition of the dummy variable *financial covenant violation* from the financial covenant violations data set by Nini et al. (2012) is presented in the Appendix: List of Abbreviations. Second, all observations with less than four consecutive previous observations are dropped.[7] This step is done in order to ensure the determination of the dummy variable that is the focus of our analysis: *New financial covenant violation*, which Nini et al. (2012, p.1724 - 1725) defines as "financial covenant violations for firms that have not violated a covenant [or granted a covenant violation waiver] in the previous four quarters". Equation (1) formalizes the variable definition for firm i at explanatory time $t = 2003Q4$ (forth quarter of 2003):

[5] S&P Capital IQ database only provides quarterly data for fiscal years.
[6] The choice of the second quarter of 1997 is determined by the second filter, explained in the following.
[7] The implementation of an algorithm taking care of this filter step was admittedly difficult.

$$\textit{New financial covenant violation}_{\,i,2003Q4} = \begin{cases} 1, & \text{if firm } i \text{ has reported or has been waived a financial covenant violation in its 10-K or 10-Q SEC filings for 2003Q4, but not for 2003Q3, 2003Q2, 2003Q1, 2002Q4.} \\ 0, & \text{otherwise} \end{cases} \tag{1}$$

Note that Nini et al. (2012) also defines the event where a covenant violation is waived as a covenant violation. As electronic SEC filings, which Nini et al. (2012) examined for the construction of their debt covenant violation dummy variable, were made mandatory from the second quarter of 1996 on, the firm's debt covenant violation data set, provided by Nini et al. (2012) starts at the second quarter of 1996. By design, the last restriction therefore sets the start of the final sample period to the second calendar quarter of 1997, as this is the earliest possible time when a financial covenant violation can be classified as either *new* or *non-new*. Ultimately, the data set on firms' accounting data is merged with the data set on firms' financial covenant violations, generating the final (baseline) sample of 174,984 firm-quarter observations from 7,075 non-financial U.S. firms. These observations range from the second (first fiscal) quarter of 1997 to the second (forth fiscal) quarter of 2009. The covenant violation data itself is available until the second (first fiscal) quarter of 2008, totaling 162,100 firm-quarters.[8]

The sample in Nini et al. (2012) consists of 7,661 firms and 181,704 firm-quarter observations from the second quarter of 1997 to the fourth quarter of 2008. The differences of the two sample sizes can be attributed to two reasons: we chose to extend the sample size by two quarters, in order to fully compensate the loss of observations due to adding variables with four leads. Furthermore, we were unable to match debt covenant violation data to accounting data for certain firms. While almost all firms from Compustat (99.88%) could be found in and matched with the debt covenant violation data set, the match of firms from S&P Capital IQ amounted only to 86.21% of all firms for which debt covenant violation data is available. Nevertheless, the relative shares of the baseline sample and the sample of Nini et al. (2012) show quite similar figures: 6.42% of firm-quarter observations (11,236 firm-quarters) include a covenant violation, which is consistent with the findings of Nini et al. (2012) (6.9%). While Nini et al. (2012) count 2.1% of their firm-quarter observations to include *new* covenant violation, the baseline sample

[8] For the sake of transparency and accountability the code to obtain the baseline data set and to compute all numbers, tables and figures that can be found in this paper was made available at `https://github.com/isabellftz/covenants`.

accounts for 1.84% (3,227 firm-quarters). Excluding the last four quarters for which no data on debt covenant violations is available in our baseline sample, the relative shares on firm-quarter level of the baseline sample are 6.93% and 1.99%, respectively, leading to results even closer to the values reported in Nini et al. (2012) (6.9% and 2.1%, respectively). The baseline sample finds 40.23% of firms (28,46 firms) to be violator firms, i.e. firms that have reported a debt covenant violation at some point over the sample period, slightly less than Nini et al. (2012) found (40,5%), while 34.28% of firms (2,425 firms) are new violator firms, i.e. have reported a *new* debt covenant violation at some point over the sample period, which amounts to roughly 85.21% of all violator firms. 10.01% of firms (708 firms) are one-time violators and 30.22% of firms (2,138 firms) are frequent violators. Figure 1 shows that between 10% and 20% (5% and 10%) of firms violated a (new) debt covenant in any given fiscal year from 1997 to 2007. Figure 1 extends Figure 2 in Nini et al. (2012) by grouping the violator firms that report a *new* debt covenant violation to either one-time or frequent violators. While "new violations follow the same cyclical pattern as total violations and reached a high of nearly 10% during [the 2001–2002 recession]" (Nini et al., 2012, p.1725), one-time violators' new violations (plotted by the intersection line of the gray areas) mostly resist the pattern.

3.2 Summary Statistics

Table 1 presents descriptive statistics for each variable in the baseline sample. The Appendix: List of Abbreviations provides variable definitions. Although Nini et al. (2012) do not explicitly mention the use of winsorizing in either their paper or the appendix, we choose to do so for all financial ratios at the bottom and top 1% levels in order to mitigate the effects of extreme outliers[9] on further descriptive and inductive statistical analysis. Negative total assets, PPENT and R&D expenditures are excluded, because, except for the event of an error in recording transaction, it is unfounded for those variables to have negative balance. Leverage and R&D scaled by total assets are between zero and one, while net worth and PPENT, both scaled by total assets, are between negative one and one, which makes sense, given that the variables in question are ratios and their respective numerator cannot exceed the denominator (in absolute values) by definition. Interest expenses and operating cash flow, both scaled by average assets, can go to infinity or (negative infinity) by construction. However, the variable definition takes into account that the two variables in question scaled by *total* assets (lagged total assets) are well defined between negative one and one. Following the variable definition of Nini et al. (2012) all

[9]Without winsorizing the maximum data point for the variables *market-to-book* and *currentratio* are at 47,289.5 and 291.8, respectively.

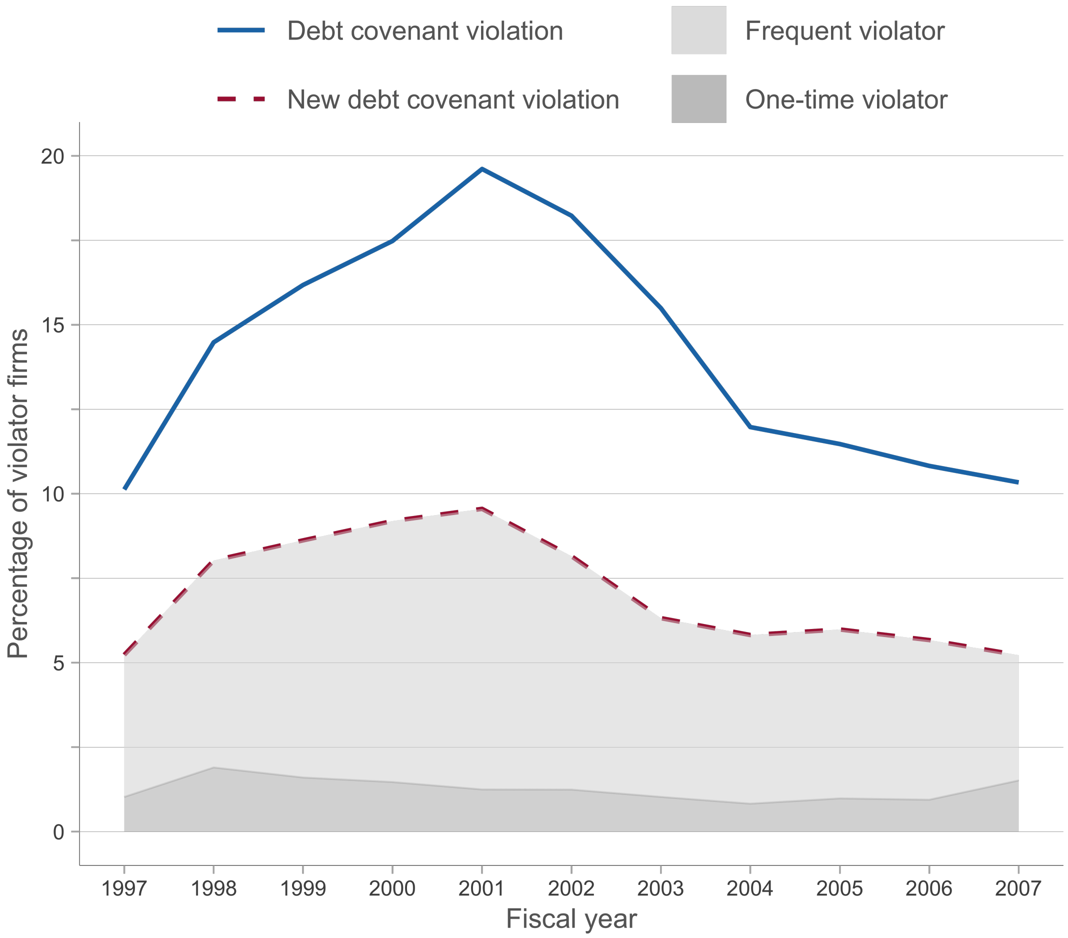


Figure 1: Financial Covenant Violations from 1997 to 2007

This figure shows the annual share of firms reporting a (new) financial covenant violation during the sample periods' fiscal years for which four quarters of covenant data were available, 1997 to 2007. The annual share of firms reporting a debt covenant violations is additionally decomposed into the annual share of frequent violator firms and one-time violator firms. Variable definitions are provided in Appendix: List of Abbreviations. Due to the four-quarter constraint on years the sample employed for this figure includes a slightly smaller number of observation, 161,857 (of 162,100) and 7,074 (of 7,075) non-financial U.S. firms. Firm data was retrieved from S&P Capital IQ database and Compustat, covenant violation data from supplementary material of Nini et al. (2012), available at `https://amirsufi.net/chronology.html`.

flow variables (operating cash flow, interest expenses and CAPEX) are annualized. Comparing the results from Table 1 and Table 2, the variables' values and ranges of the baseline sample match those provided by Nini et al. (2012) quite precisely. The four firm-quarter changes in log assets and log PPENT are slightly higher than their counterpart values in Nini et al. (2012), rooted in, on average, marginally higher levels of total assets and PPENT in the baseline sample.

Comparing descriptive statistic for market-to-book ratio, values for statistical measures, which are moderately robust to outliers (median, $10th$ percentile and $90th$ percentile) of the baseline sample, appear to be coherent with the values found in Nini et al. (2012). Mean values for operating cash flow scaled by average assets and market-to-book ratio, as well as standard deviation values for the former variable and current ratio are slightly distorted by some outliers, if we take the figures in Nini et al. (2012) as a base reference. As mentioned previously, the two data sets exhibit very similar characteristics, with most comparable variables lying in similar ranges. Covering almost identical time periods, hence resulting in two homogeneous economic backgrounds, one can argue that the baseline sample can serve as successful replication of the sample used in Nini et al. (2012). Further analysis conducted on the basis of the baseline sample can therefore be seen as an extension of the analysis performed by Nini et al. (2012).

Table 1: Summary Statistics

	No. of obs.	Mean	SD	Min	10^{th}	Median	90^{th}	Max	No. of firms
Operating cash flow/ Average assets	173,330	0.040	0.281	−1.445	−0.248	0.101	0.260	0.591	7,057
Leverage	165,919	0.224	0.213	0.000	0.000	0.186	0.526	0.871	6,994
Interest expense/ Average assets	143,909	0.026	0.041	0.000	0.000	0.016	0.056	0.317	6,551
Net worth/ Assets	171,885	0.488	0.283	−0.543	0.153	0.504	0.837	0.961	7,020
Current ratio	174,862	2.939	3.349	0.038	0.769	1.960	5.830	24.600	7,074
Market-to-book	162,704	2.705	4.550	0.180	0.662	1.411	5.147	37.687	7,029
Change in Ln(Assets)	151,059	0.063	0.449	−8.990	−0.266	0.046	0.402	13.078	6,633
Change in Ln(PPENT)	149,429	0.044	0.524	−10.792	−0.312	0.026	0.437	10.463	6,591
CAPEX/ Average assets	168,225	0.059	0.095	−2.247	0.004	0.033	0.135	2.412	7,035
R&D/ Assets	87,185	0.033	0.057	0.000	0.000	0.018	0.078	1.000	4,384
Assets ($ millions)	174,984	2,290	14,533	0.001	17.100	196.800	3,465	829,550	7,075
PPENT ($ millions)	173,450	761.967	3,933	0.001	1.470	35.700	1,086	127,861	7,041
Assets (4 leads)/ Assets	156,769	10.021	1,936	0.000	0.763	1.044	1.484	478,100	6,634
PPENT (4 leads)/ PPENT	155,044	1.971	104.133	0.000	0.731	1.024	1.539	35,000	6,598
PPENT/ Assets	173,449	0.269	0.233	0.000	0.038	0.193	0.645	1.000	7,041
Financial covenant violation	162,100	0.069	0.254	0.000	0.000	0.000	0.000	1.000	7,075
New financial covenant violation	162,100	0.020	0.140	0.000	0.000	0.000	0.000	1.000	7,075
One-time violator	162,100	0.102	0.302	0.000	0.000	0.000	1.000	1.000	7,075
Frequent violator	162,100	0.358	0.480	0.000	0.000	0.000	1.000	1.000	7,075

This table presents number of observations, mean, standard deviation (SD), minimum (Min), $10th$ percentile, median, $90th$ percentile, maximum (Max) and number of firms for each variable of the sample employed in this paper. This sample consists of 174,984 firm-quarter observations from 7,075 non-financial U.S. firms from the second (first fiscal) quarter of 1997 to the second (forth fiscal) quarter of 2009 with covenant violation data available until the second quarter of 2008 (162,100 firm-quarters). Firm data was retrieved from S&P Capital IQ database and Compustat database, covenant violation data from supplementary material of Nini et al. (2012), available at `https://amirsufi.net/chronology.html`. Variable definitions are presented in the Appendix: List of Abbreviations. All financial ratios are winsorized at 1% and 99% levels of their distribution.

Table 2: Summary Statistics in Nini et al. (2012, Table 5, p. 1735)

	No. of obs.	Mean	SD	10^{th}	Median	90^{th}
Operating cash flow/ Average assets	179,350	0.057	0.237	−0.200	0.104	0.260
Leverage	176,291	0.242	0.242	0.000	0.197	0.549
Interest expense/ Average assets	172,190	0.021	0.028	0.000	0.013	0.050
Net worth/ Assets	181,698	0.475	0.306	0.145	0.498	0.830
Current ratio	177,749	2.787	2.783	0.818	1.950	5.482
Market-to-book	181,698	2.027	1.775	0.850	1.442	3.764
Change in Ln(Assets)	165,968	0.057	0.380	−0.258	0.041	0.381
Change in Ln(PPENT)	164,990	0.038	0.484	−0.299	0.023	0.418
CAPEX/ Average assets	177,014	0.059	0.078	0.005	0.034	0.138

This table provides descriptive statistics for selected variables for the sample used by Nini et al. (2012), which covers 7,661 non-financial U.S. firms and 181,704 firm-quarter observations from the second quarter of 1997 to the fourth quarter of 2008. All values are transcribed from Nini et al. (2012, Table 5, p.1735). Minimum and maximum values for each variable are not provided in Nini et al. (2012)

4 Methodology and Results

To study the influence of creditors on firm investment policy, one-year changes in four measures of firm investment behavior, namely total assets (logarithmized), PPENT (logarithmized)[10], CAPEX (scaled by total assets) and R&D (scaled by total assets), following a new financial covenant violation[11] are examined. The definitional design of a new debt covenant violation in Equation (1) allows us to isolate the impact of debt covenant violations on the four outcome variables, as it ensures a debt covenant violation that is reported for firm i at time t is not preceded by a violation in the previous four quarters. In order to test for statistical significance of post-violation changes in firm investment policy, two commonly used variants of panel data analysis are employed: a pooled OLS regression analysis and a fixed-effects regression analysis. Pooled regression analysis is used to estimate the cumulative effect of a new financial covenant violation on the four investment measures, one year after the new violation was reported, whereby the assumptions hold that there are no time or industry fixed effects on the four outcome variables. With these preliminary findings in mind, this assumptions is relaxed and possible dynamics within industries are examined. This effect is captured by applying a linear fixed effects panel regression model. Based on this underlying methodology, the two hypotheses derived in Section 2 are assessed. Table 3 reports estimates for all models using the aforementioned four measures of firms' investment behavior as the dependent variable.

[10] For expositional ease, we will refer to logarithmized variables as *log variables*.

[11] Four quarter (one-year) changes in any investment measure after a new debt covenant violation was reported are often referred to simply as *post-violation* changes in the following.

Table 3: Regression of Investment Measures
Panel A

	Change in Ln(Assets)				Change in Ln(PPENT)			
(Column no.) Model	(1) I	(2) II	(3) III	(4) IV	(5) I	(6) II	(7) III	(8) IV
New financial covenant violation (NFCV)	−0.092***	−0.046***	−0.041***	−0.036***	−0.100***	−0.036***	−0.030***	−0.029***
	(0.008)	(0.008)	(0.008)	(0.008)	(0.010)	(0.010)	(0.010)	(0.011)
Marginal effect of NFCV (at mean in $ millions)	−0.922	−0.460	−0.410	−0.360	−0.197	−0.071	−0.059	−0.057
One-time violator (OTV)				0.017**				0.022**
				(0.007)				(0.009)
Frequent violator (FV)				−0.016***				−0.006
				(0.006)				(0.007)
Operating cash flow/ Average assets		0.194***	0.200***	0.202***		0.201***	0.204***	0.205***
		(0.005)	(0.032)	(0.032)		(0.006)	(0.025)	(0.025)
Leverage		−0.079***	−0.060**	−0.057**		0.022*	0.060**	0.061**
		(0.009)	(0.023)	(0.023)		(0.012)	(0.029)	(0.030)
Interest expense/ Average assets		0.313***	0.347*	0.350*		0.680***	0.629**	0.629**
		(0.054)	(0.183)	(0.182)		(0.071)	(0.258)	(0.258)
Net worth/ Assets		0.078***	0.093***	0.091***		0.179***	0.217***	0.215***
		(0.007)	(0.022)	(0.022)		(0.010)	(0.027)	(0.027)
Current ratio		−0.004***	−0.005***	−0.005***		0.008***	0.007***	0.007***
		(0.000)	(0.001)	(0.001)		(0.001)	(0.002)	(0.002)
Market-to-book ratio		0.019***	0.018***	0.018***		0.015***	0.014***	0.014***
		(0.000)	(0.001)	(0.001)		(0.000)	(0.001)	(0.001)
Constant	0.065***	−0.008			0.046***	−0.156***		
	(0.001)	(0.006)			(0.001)	(0.008)		
Firm controls	No	Yes	Yes	Yes	No	Yes	Yes	Yes
FE & Clustered SE	No	No	Yes	Yes	No	No	Yes	Yes
No. of obs.	151.059	108.204	108.204	108.204	149.429	107.946	107.946	107.946
No. of firms	6.633	5.885	5.885	5.885	6.591	5.878	5.878	5.878
Adj. R^2	0.001	0.079	0.110	0.111	0.001	0.067	0.093	0.093

(continued)

Table 3 (Continued)
Panel B

	Change in CAPEX/ Average assets				Change in R&D/ Assets			
(Column no.) Model	(9) I	(10) II	(11) III	(12) IV	(13) I	(14) II	(15) III	(16) IV
New financial covenant violation (NFCV)	−0.009***	−0.007***	−0.006***	−0.006***	−0.003**	−0.001	−0.001	−0.001
	(0.002)	(0.002)	(0.002)	(0.002)	(0.001)	(0.001)	(0.001)	(0.001)
One-time violator (OTV)				0.002**				0.000
				(0.001)				(0.001)
Frequent violator (FV)				0.000				0.000
				(0.000)				(0.001)
Operating cash flow/ Average assets		0.012***	0.013***	0.013***		0.005***	0.006*	0.006*
		(0.001)	(0.002)	(0.002)		(0.001)	(0.003)	(0.003)
Leverage		−0.013***	−0.009*	−0.009*		0.006***	0.008***	0.008***
		(0.002)	(0.004)	(0.004)		(0.001)	(0.003)	(0.003)
Interest expense/ Average assets		0.069***	0.080***	0.080***		0.016**	0.016	0.016
		(0.014)	(0.028)	(0.028)		(0.007)	(0.021)	(0.021)
Net worth/ Assets		−0.008***	−0.003	−0.003		0.004***	0.005*	0.005*
		(0.002)	(0.002)	(0.002)		(0.001)	(0.003)	(0.003)
Current ratio		0.001***	0.001***	0.001***		0.001***	0.001***	0.001***
		(0.000)	(0.000)	(0.000)		(0.000)	(0.000)	(0.000)
Market-to-book ratio		0.000*	0.000	0.000		0.000***	0.000***	0.000***
		(0.000)	(0.000)	(0.000)		(0.000)	(0.000)	(0.000)
Constant	−0.004***	0.002			0.000**	−0.003***		
	(0.000)	(0.002)			(0.000)	(0.001)		
Firm controls	No	Yes	Yes	Yes	No	Yes	Yes	Yes
FE & Clustered SE	No	No	Yes	Yes	No	No	Yes	Yes
Num. obs.	141.434	101.035	101.035	101.035	73.797	48.906	48.906	48.906
No. of firms	6.556	5.810	5.810	5.810	4.031	3.473	3.473	3.473
Adj. R^2	0.000	0.048	0.052	0.052	0.000	0.015	0.020	0.020

This table provides estimates of the Model I (columns (1), (5), (9), (13)) and II ((2), (6), (10), (14)), using OLS, and of Model III ((3), (7), (11), (15)) and IV ((4), (8), (12), (16)), using and linear fixed effects. We employ four dependent variables to proxy for four-quarter changes in firms' financial investment policy observed after a new financial covenant violation: changes in total assets (logarithmized) and property, plant, and equipment (PPENT) (logarithmized) are examined in Panel A, Changes in capital expenditures (scaled by average assets) and research and development expenditure (R&D) (scaled by total assets) are examined in Panel B. All regression models include a set of covenant control variables (operating cash flow scaled by average assets; the leverage ratio; the ratio of interest expense to average assets; the ratio of net worth to total assets; the current ratio; and the market-to-book ratio). If stated, specifications include controls for firm characteristics (firm controls) (the level and first difference of log assets; the level and first difference of the ratio of PPENT to total assets), industry, fiscal-quarter and year-quarter fixed effects (FE) and clustered standard errors (SE) by firm and year-quarter. Estimates of firm controls can be found in raw table output on GitHub in output/tables/Table3.pdf. Variable definitions are provided in Appendix: List of Abbreviations. Standard errors are reported in parentheses. *, **, and *** denote statistical significance at 10%, 5%, and 1% level. The full sample used for estimation of all models consists of 174,984 firm-quarter observations of 7,075 non-financial U.S. firms from 1997 to 2009 (starting and ending with the second quarter), with covenant violation data available until the second quarter of 2008. Firm data was retrieved from S&P Capital IQ database and Compustat, covenant violation data from supplementary material of Nini et al. (2012), available at `https://amirsufi.net/chronology.html`.

4.1 Assessing Hypothesis I

In order to empirically investigate the hypothesized differential investment responses, to new debt covenant violations, we assume that four-quarter changes in the four investment measures for firm i at time t, $Y_{i,t+4} - Y_{i,t}$, can each be modeled as:

$$Y_{i,t+4} - Y_{i,t} = \gamma_0 + \gamma_1 NFCV_{i,t} + \mu_{i,t} \tag{I}$$

where i indexes firm $i = 1, ..., N$ with N, denoting the individual number of firms used in the four regression estimations, for which results are provided in Table 3. t indexes time t=1997Q2, 1997Q3, 1997Q4, 1998Q1, ..., 2008Q2 with Q denoting the respective quarter of the stated year. Indexes' definition apply to all further refinement of Model I. $NFCV_{i,t}$ is a dummy variable which equals one if firm i at time t reports a new debt covenant violation and zero otherwise. Given the post-violation changes of log assets are to be estimated, the right hand side of the equation in I yields $Ln(Assets_{i,t+4}) - Ln(Assets_{i,t}) = Ln(\frac{Assets_{i,t+4}}{Assets_{i,t}})$. The estimated coefficient for $NFCV_{i,t}$, γ_1, is a semi-elasticity and thus is interpreted as percentage change in the four-quarter changes (growth rate) of total assets. To further illustrate the interpretation, Panel A of Table 3 also lists the marginal effects of $NFCV_{i,t}$, evaluated at the respective baseline sample average, here $\frac{Assets_{i,t+4}}{Assets_{i,t}}$, and formalized in equation 2:

$$\frac{\partial \frac{Assets_{i,t+4}}{Assets_{i,t}}}{\partial NFCV} = \gamma_1 \, \mathbb{E}(\frac{Assets_{i,t+4}}{Assets_{i,t}})^{12} = -0.092 \times 10.021 \approx -0.922 \ (\$ \ millions) \tag{2}$$

Equation (2) analogously applies to four-quarter changes of log PPENT as dependent variable in Model I. As can be inferred from columns (1), (5), (9) and (13) of Table 3, a new debt covenant violation decreases the growth rate or level of all four measures of investment in the year following the violation. For instance, column (1) of Table 3 indicates that total assets growth of firms violating a debt covenant falls, on average, by 9.2% post-violation, c.p., implying approximately an average \$922,000 decrease in total assets, evaluated at the sample average of changes in total assets, one year after a debt covenant violation had been reported. Column (13) of Table 3 indicates an, on average, decline of R&D-to-assets ratio by 0.3 percentage points, c.p., post-violation. In all four cases, the estimated effect is statistically significant; however, this parsimonious modelling approach is arguably not sufficient in order to test hypothesis I, as it does not account for confounding factors that may have an independent effect on the four outcome variables. Therefore, our estimates fail to be economically important. Attempting to

[12] Equals the mean of the variable *Assets (4 leads)/ Assets*, cf. Table 1.

correct for that, we add a set of common debt covenant-specific characteristics $q_{i,t}$ and a set of common firm characteristics $p_{i,t}$ to Model I. Hence, the second model to be estimated is:

$$Y_{i,t+4} - Y_{i,t} = \gamma_0 + \gamma_1 NFCV_{i,t} + \alpha' q_{i,t} + \beta' p_{i,t} + \mu_{i,t} \tag{II}$$

where i indexes firm, t indexes time. The variable definitions from Model I hold. Following the model specification in Nini et al. (2012), $q_{i,t}$ is a vector that includes five covenant controls (operating cash flow-to-average assets, leverage, interest expense-to-average assets, net worth-to-assets, current ratio) and market-to-book ratio. The choice of the former five is based on Roberts and Sufi (2009) findings, which show that cash flow-based ratios, debt-to-total assets ratio, coverage ratios, net worth-to-tangible net worth ratios and liquidity-based ratios are the most commonly used financial ratios, on which debt covenants are written in loan agreements. Market-to-book ratio is included as "powerful" (Nini et al., 2012, p. 1734), as well as fairly common predictor in empirical studies that examine the impact of debt covenant violations on firm policies (Ferreira et al. (2018), Roberts and Sufi (2009), Rajan and Zingales (1995)). Following the model specification of Nini et al. (2012), $p_{i,t}$ is a vector of variables that control for four firm characteristics: the level and first difference of log assets and the level and first differences of PPENT scaled by total assets. Columns (2), (6), (10), (14) of Table 3 show, the new subset of explanatory variables increases the model's explanatory power in each panel, by 7.8%, 6.6%, 4.8% and 1.5%, respectively, compared to Model I. All added control variables are statistically significant for each panel specification and therefore of importance in the model. Consistent with the results of Chava and Roberts (2008), operating cash flow (scaled by average assets) is positively, while leverage is negatively related to the ratio of CAPEX to average assets. In fact, the former positive relationship applies to all four investment measures. Over a period of four quarters, c.p., a one percentage point increase in the ratio of operating cash flow to average assets is associated with a \$1.9 and \$0.4 million increase of total assets and PPENT, and a 1.2 and 0.5 percentage points increase of CAPEX-to-average assets ratio and R&D-to-assets ratio, respectively. As firms' leverage increases, c.p., firms on average indicate a lower ratio of CAPEX to average assets and asset growth, but a higher ratio of R&D to total assets and PPENT growth. This makes sense as former studies on firms' capital structure (Cvijanović (2014) and Hall (2012)) have shown that tangibility, measured by PPENT-to-assets ratio, is on average positively related to leverage. Four quarters post-violation, the estimated negative impact of new covenant violations on total assets, PPENT and CAPEX-to-average assets ratio is lowered by, but robust to the inclusion of the control variables and remains statistically significant. However, we find negative, albeit insignificant post-violation changes in R&D-to-assets ratio.

Yet, the specifications in Model II exhibits some shortcomings. Employing panel data and four-quarter differences as dependent variables, our observations within each firm and year-quarter are not independently and identically distributed, thus inducing biased idiosyncratic standard errors. Moreover, estimating our model via a pooled OLS method only to explains the variation in average four-quarter changes of the four investment measures *across* all industries and time periods. Unobserved time-invariant, industry-specific effects, seasonal patterns related to fiscal quarters, or industry-invariant possible macroeconomic annual alterations are not controlled for, and so unobserved heterogeneity, a consequence of omitted variables, can distort measurements of our estimates. There are several reasons for believing investment may vary across industries. Titman and Wessels (1988) argued that firms with relatively high R&D expenditures have lower debt ratios on average, while Lang et al. (1996) found a negative relationship between leverage and CAPEX-to-fixed assets for firms in industries with low growth opportunities. With the results of Bradley et al. (1984), which indicated that the average firm leverage depends on the firm's industry, we conclude that firms in highly leveraged industries with few growth opportunities, e.g. the steel industry, employ less R&D expenditures and have a lower ratio of CAPEX-to-fixed assets than firms in the less leveraged industries, like the drugs and cosmetics industry. A common approach to mitigating concerns about time- and industry-invariant omitted variables is to include dummy variables for each industry and time period to absorb all across-industry and -time effects. As done similarly by Nini et al. (2012), we capture the variation *within* each year-quarter, fiscal quarter and industry by estimating the following fixed effects specification:

$$
\begin{aligned}
Y_{i,t+4} - Y_{i,t} &= \rho_t + \theta_{i,t} + \zeta_i + \gamma_1 NFCV_{i,t} + \alpha' q_{i,t} + \beta' p_{i,t} + \epsilon_{i,t} \ , \qquad \text{(III)} \\
with \qquad \rho_t &= \sum_{t=1997Q3}^{2008Q2} \phi_t CQY_t, \\
\theta_{i,t} &= \sum_{f=2}^{4} \delta_f FQ_{f,i,t} \\
\zeta_i &= \sum_{\substack{k=2 \\ k \notin \{32,35\}}}^{38} \lambda_k FFI_{k,i}
\end{aligned}
$$

where i indexes firm and t indexes time. The variable definitions from Model I and Model II hold. To ensure our inference is robust to heteroskedasticity and autocorrelation, standard errors in Model III are clustered by firm and calendar year-quarters. CQY_t represents (45-1) year-quarter dummy variables, $FQ_{f,i,t}$ represents (4-1) fiscal quarter dummy variables, and $FFI_{k,i}$ represents the (38-2-1) Fama and French industry classification dummy variables, derived from firms' standard industrial classification (SIC) codes[13]. To avoid perfect multicollinearity, one dummy of each dummy group is omitted from Model III. We do not find any firms in our baseline sample that operate in the irrigation systems industry (industry 32), nor in the finance, insurance, and real estate industry (industry 35), as the latter was actively excluded from the baseline sample in the sample construction process. Therefore, the industry dummies $FFI_{32,i}$ and $FFI_{35,i}$ are dropped. Consider as an example time $t = 2003Q4$, firm i that operates in the service industry, industry 36, and designates December as its fiscal year-end month, we get $CQY_{2003Q4} = 1$, $FQ_{4,i,2003Q4} = 1$, $FFI_{36,i} = 1$, while all other dummy variables in the model are zero. We can describe the adjustment process from Model II to Model III by $\mu_{i,t} = \rho_t + \theta_{i,t}$ $+ \zeta_i + \epsilon_{i,t}$. Industry-fixed effects, ζ_i is assumed to have a covariance unequal to zero with either of our regressors. This assumption is likely to hold because, as the results of Model III (columns (3), (7), (11), (15)) compared to Model II in Table 3 show, conditioning on industry fixed effects significantly decreases our estimators for $NDCV$ in all panels, expect for the panel with R&D-to-assets ratio as dependent variable. Table 4 compares the estimates for four-quarter changes of log assets, log PPENT and CAPEX (scaled by total assets) of Model III to the corresponding estimates reported by Nini et al. (2012). Overall, our results show a high degree of similarity. We find that the growth rate of total assets, PPENT and ratio of CAPEX to average assets is on average and c.p., 4.1%, 3% and 0.06 percentage points lower at time t for firms that have reported a new debt covenant four quarters prior (at time $t - 4$), than for firms with no such violation report, with the difference statistically significant at 1% level. The corresponding estimated effect sizes in Nini et al. (2012) are 4.1%, 4.2% and 0.05 percentage points at 1%, 1% and 5% significance level, respectively. Both studies detect evidence that the event of a new debt covenant is statistically significant in explaining firms' decrease in the first three investment measures; however, the value of adjusted R^2 indicates that only about 11% (vs. 17% in Nini et al. (2012)), 9% (vs. 14%) and 5% (vs. 6%) of the variation in firm debt ratio is explained by the Model III. The estimated coefficients for market-to-book ratio in our analysis discernibly differ in their magnitude from the ones obtained by Nini et al. (2012). The reason may be that the average market-to-book value in our analysis is approximately 0.39 market-to-book standard deviations larger than the average market-to-book value in Nini et al. (2012). Our penultimate results of Model III support Hypothesis I for the most part. Only the estimated negative impact of new debt covenant violation on the ratio of R&D expenditures-to-assets did not prove to be significant post-violation.

[13] Available at https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/Data_Library/det_38_ind_port.html for download.

Table 4: Comparison of Regression Results

	Change in Ln(Assets)		Change in Ln(PPENT)		Change in CAPEX/ Average Assets	
obtained from (column no.)	Model III Table 3.A (3)	Nini et al. (2012) Table 6.A (1)	Model III Table 3.A (7)	Nini et al. (2012) Table 6.A (5)	Model III Table 3.B (11)	Nini et al. (2012) Table 6.B (9)
New financial covenant violation (NFCV)	−0.041***	−0.041***	−0.030***	−0.042***	−0.006***	−0.005**
	(0.008)	(0.007)	(0.010)	(0.011)	(0.002)	(0.002)
Operating cash flow/ Average assets	0.200***	0.309***	0.204***	0.312***	0.013***	0.011***
	(0.032)	(0.030)	(0.025)	(0.022)	(0.002)	(0.002)
Leverage	−0.060**	−0.095***	0.060**	−0.024	−0.009*	−0.007**
	(0.023)	(0.023)	(0.029)	(0.029)	(0.004)	(0.003)
Interest expense/ Average assets	0.347*	0.604***	0.629**	0.785***	0.080***	0.122***
	(0.183)	(0.208)	(0.258)	(0.240)	(0.028)	(0.021)
Net worth/ Assets	0.093***	0.031	0.217***	0.110***	−0.003	0.001
	(0.022)	(0.023)	(0.027)	(0.023)	(0.002)	(0.001)
Current ratio	−0.005***	−0.005***	0.007***	0.006***	0.001***	0.001***
	(0.001)	(0.001)	(0.002)	(0.001)	(0.000)	(0.000)
Market-to-book ratio	0.018***	0.063***	0.014***	0.047***	0.000	−0.001***
	(0.001)	(0.003)	(0.001)	(0.002)	(0.000)	(0.000)
Firm controls	Yes	Yes	Yes	Yes	Yes	Yes
FE & Clustered SE	Yes	Yes	Yes	Yes	Yes	Yes
No. of obs.	108.204	150.474	107.946	150.019	101.035	145.442
Adj. R^2	0.11	0.17	0.09	0.14	0.05	0.06

This table shows a comparison of the fixed effects panel regression results from Model III and Table 6, column (1), (5) and (9) in Nini et al. (2012, p. 1738-1739), each estimating the effects of a new debt covenant violation on three measures of firms' investment policies one year post-violation: growth rate of total assets and PPENT and level of capital expenditures (scaled by average assets). Variable definitions are provided in Appendix: List of Abbreviations. All regression models include two sets of control variables: covenant controls (for which all estimates are presented) and controls for firm characteristics (firm controls), which are the level and first difference of total log assets and the level and first difference of the ratio of PPENT to total assets. Estimates for firm controls were not published by Nini et al. (2012). For Model III they can be found in raw table output on GitHub in output/tables/Table3.pdf. All specifications include industry, fiscal-quarter and year-quarter fixed effects (FE). Standard errors (SE) are clustered on firm and year-quarter level and are reported in parentheses. Levels of significance are indicated as follows: $*p < 0.1$, $**p < 0.05$, $***p < 0.01$. The full sample used for estimation of Model III consists of 174,984 firm-quarter observations of 7,075 non-financial U.S. firms from 1997 to 2009 (starting and ending with the second quarter), with covenant violation data available until the second quarter of 2008. Firm data was retrieved from S&P Capital IQ database and Compustat, covenant violation data from supplementary material of Nini et al. (2012), available at `https://amirsufi.net/chronology.html`. For number of firms refer to Table 3. The full sample used for estimation in Nini et al. (2012) consists of 181,704 firm-quarter observations sample of 7,661 non-financial U.S. firms from the second quarter of 1997 to the fourth quarter of 2008.

4.2 Assessing Hypothesis II

In our attempts to assess Hypothesis II, we ask whether the two groups exhibits significant differences in debt covenants-specific control variables and all four investment measures. Table 5 provides univariate comparisons of one-time violators ($j = 1$) and frequent violators ($j = 2$). We present mean values for each variable in each group of violators and the differences in mean values between the two groups. To determine if the group mean difference of each variable reported in Table 5, x, is significant, we test the null hypothesis $H_0 : \overline{x}_1 = \overline{x}_2$ (while $H_1 : \overline{x}_1 \neq \overline{x}_2$), where $\overline{x}_j$ with $j \in \{1, 2\}$ denotes the mean of variable x from group j. Since our year-quarter observations are not independent over time for each group, the Welch t-test test statistics and the respective p-values[14] are based on standard errors clustered by firm to avoid overstating the degrees of freedom.

Table 5: Averages for One-Time Violator and Frequent Violator Groups

	One-Time Violator (1)	Frequent Violator (2)	Difference (1)−(2)	t-Statistic
Operating cash flow/ Average assets	0.070	0.051	0.019**	2.298
Leverage	0.237	0.276	−0.039***	−4.524
Interest expense/ Average assets	0.027	0.032	−0.006***	−4.123
Net worth/ Assets	0.485	0.422	0.064***	5.766
Current ratio	2.664	2.309	0.356***	3.313
Market-to-book	2.509	1.881	0.627***	4.257
Change in Ln(Assets)	0.085	0.033	0.052***	6.585
Change in Ln(PPENT)	0.070	0.008	0.062***	6.730
CAPEX/ Average assets	0.067	0.059	0.008**	2.538
R&D/ Assets	0.027	0.025	0.002	0.764
Assets ($ millions)	1,334	972.180	361.374	1.757
PPENT ($ millions)	431.406	378.614	52.792	0.668
Assets (4 leads)/ Assets	1.196	25.582	−24.386	−1.003
PPENT (4 leads)/ PPENT	2.091	1.740	0.351	0.391
PPENT/ Assets	0.289	0.276	0.012	1.049
No. of obs.	16.469	58.112		
No. of firms	708	2,138		

This table presents sample averages of the most common financial covenant ratios and firm characteristics for observations of one-time violator firms and frequent violator firms. One-time violators (Frequent violators) are firms that reported one (more than one) debt covenant violation during the second quarter of 1997 and the second quarter of 2008. The sample on which the group samples are derived from consists of 174,984 firm-quarter observations of 7,075 non-financial U.S. firms from 1997 to 2009 (starting and ending with the second quarter).

[14] p-values are presented in the raw table output hosted on GitHub in output/tables/Table5.pdf.

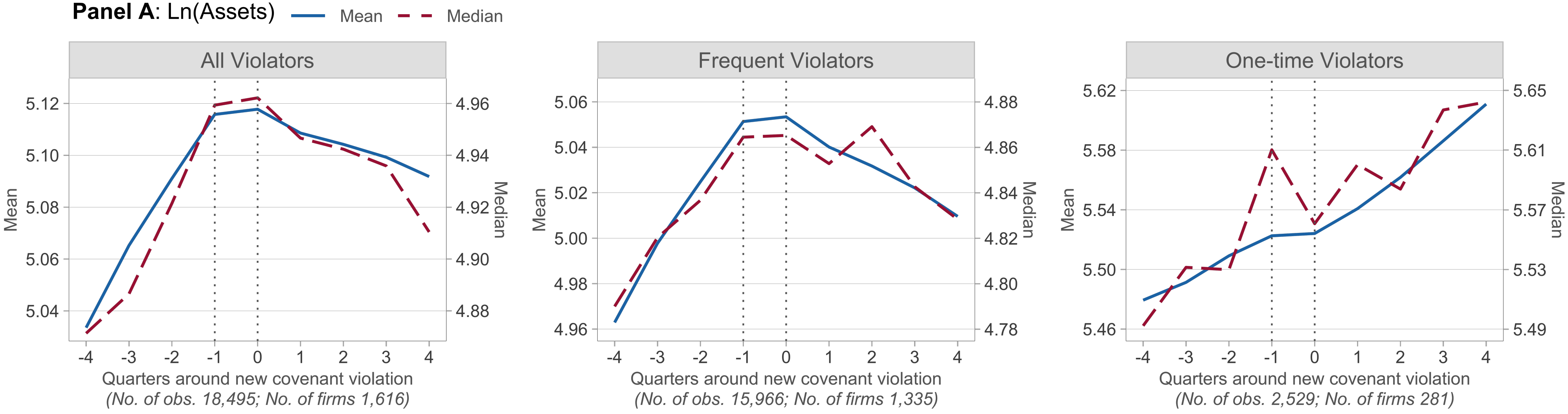
The clustered standard errors are obtained by fitting variable x_j, $j \in \{1,2\}$, in a linear regression model on a single constant, using OLS with clustered standard errors on firm level. We run this regression for each variable in each group. As can be inferred from Table 5, frequent violators have significantly lower ratios of operating cash flow to average assets, current assets to current liabilities, net worth to assets and market to book than one-time violators. In addition they have higher ratios of interest-expenses-to-average assets and debt than one-time violators. Ferreira et al. (2018) calls the last finding a "mechanical result, [as] leverage directly affects the variable that defines a violation"(p. 2398). Indeed, group mean differences of all five covenant ratio are significant at 1% and 5% level. One-time violators have a statistically significantly higher growth rate in total assets and PPENT, as well as level of CAPEX-to-average assets than frequent violators. There are no statistically significant differences in the ratio of R&D-to-assets. The last four magnitudes of our group means are confirmed in Figure 2. Panels A to D of Figure 2 present the mean and median values for our four investment measures for all, frequent and one-time violator firms' four quarters, i.e. one year before[15] and after a new debt covenant violation. More precisely, we denote by $q = 0$ the quarter in which the new debt covenant violation was reported, hence the violation occurs between $q = -1$ and $q = 0$ with the respective time interval being approximately one quarter. In order to clearly identify the effect of a new debt covenant violation on a firm's investment behaviour, we require that a firm has available four year-quarter observations pre- and post-violation, but allow violations to overlap.[16] However, the results are qualitatively similar if we exclude overlapping year-quarter observations. The evolution of log assets, log PPENT, CAPEX scaled by average assets and R&D expenditures scaled by total assets in Panel A, B, C and D of Figure 2, respectively, for *all violators* confirms the direction and approximate size of our estimated coefficients from Model III. For instance, for all violators the median capital expenditures falls from 3.3% of average assets in $q = 0$ to approximately 2.7% of average assets in $q = 4$, while the difference of -0.6 equals our estimated effect, $\hat{\gamma}_1$. Each plot in Panel A, B and C for all violators is further consistent with the findings of the graphical analysis in Nini et al. (2012). Figure 2 extends the graphical analysis in Nini et al. (2012) by dividing the data set of all violators into two groups, frequent violators and one-time violators, so that we can graphically analyze the trend of each investment measure for each group separately. Due to our restriction on firms' year-quarter observations number in Figure 2, the graphical analysis is isolated to smaller sub-samples. We therefore also choose to estimate

[15] In the following referred to as *pre-violation*.

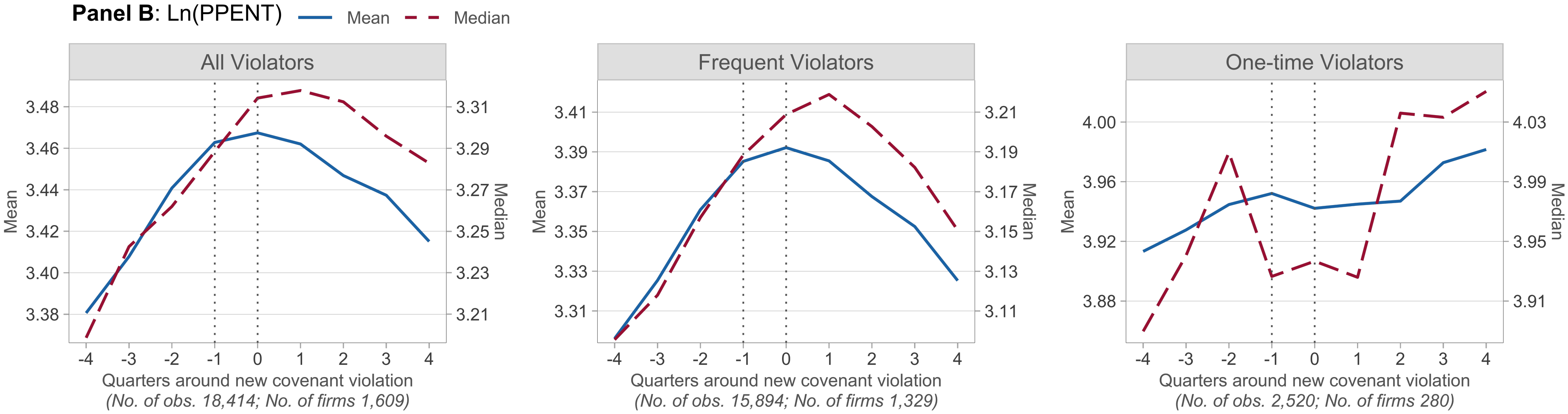
[16] E.g. a firm reports a new debt covenant in 2003Q4 and 2005Q1 and has no missing data between 2002Q4 and 2006Q1 for the variable in question, we include two sets of nine year-quarter observations: 2002Q4-2004Q4 and 2004Q1-2006Q1.

Figure 2: Firms' Investment Decisions around New Debt Covenant Violations

Panel A: Ln(Assets)



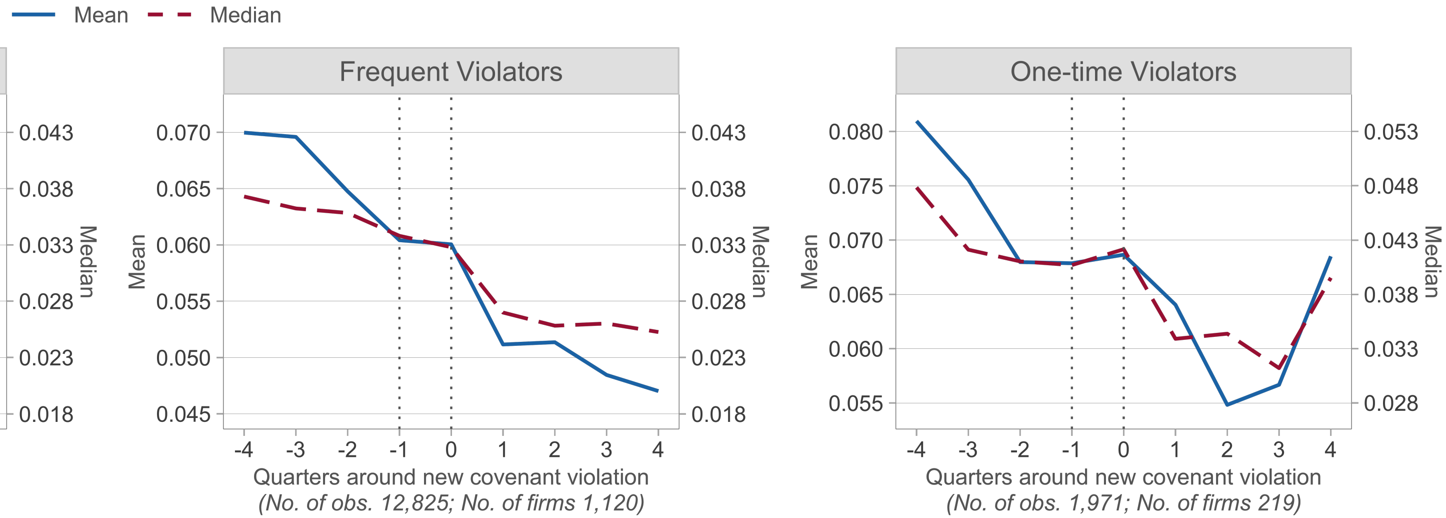
Panel B: Ln(PPENT)



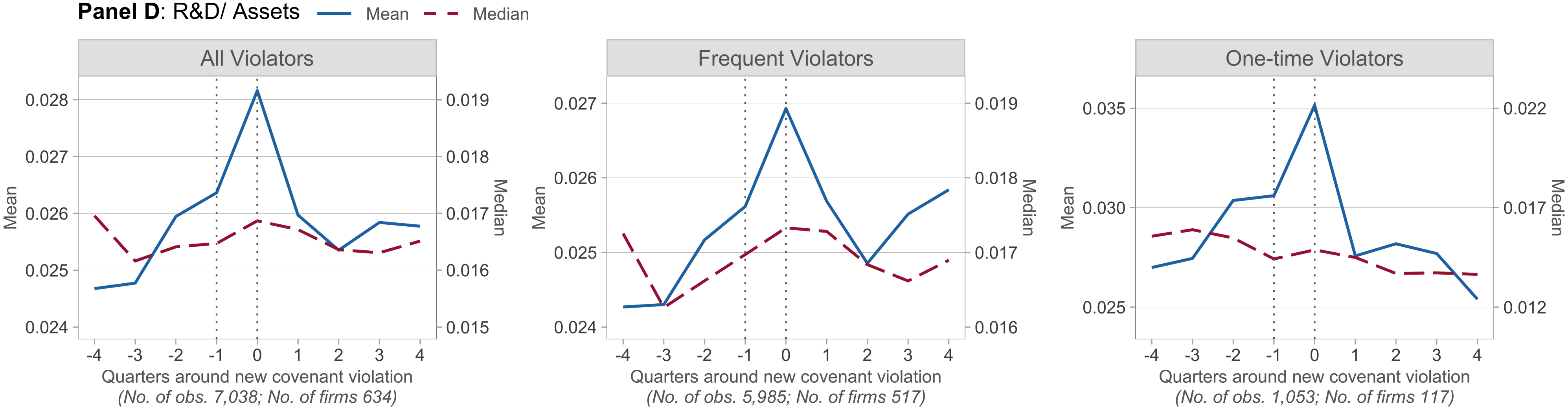
(continued)

Figure 2 (Continued)

Panel C: CAPEX/ Average Assets



Panel D: R&D/ Assets



This figure presents means and medians for four measures of investment around a violation of a new debt covenant: Total assets (logarithmized) in Panel A, PPENT (logarithmized) in Panel B, CAPEX (scaled by average assets) in Panel C and R&D (scaled by total assets) in Panel D. Variable definitions are provided in Appendix: List of Abbreviations. As a debt covenant violation is first reported at quarter 0, one can derive it took place at some point between quarter 1 and 0. Only firms with complete observations, four quarter pre- and post-violation, of the initial sample of 174,984 firm-quarter observations of 7,075 non-financial U.S. firms from 1997 to 2009 (starting and ending with the second quarter) are included in the sample used for this figure. The concrete number of observations and number of firms are stated in parentheses below each plot. Firm data was retrieved from S&P Capital IQ database and Compustat, covenant violation data from supplementary material of Nini et al. (2012), available at `https://amirsufi.net/chronology.html`.

the effect by the following specification:

$$Y_{i,t+4} - Y_{i,t} = \rho_t + \theta_{i,t} + \zeta_i + \gamma_1 NFCV_{i,t} + \gamma_2 FV_i + \gamma_3 OTV_i + \alpha' x_{i,t} + \beta' p_{i,t} + \epsilon_{i,t} \qquad \text{(IV)}$$

where i indexes firm and t indexes time. The previous variable definitions from Model I, II and III hold. Standard errors are clustered by firm and calendar year-quarters. FV_i (OTV_i) is a dummy variable which equals one if firm i is identified as frequent violator (one-time violator) and zero otherwise. If both variables take zero as their value, firm i is a non-violator. Columns (4), (8), (12) and (16) of Table 3 present the results.

Panel A of Figure 2 reveals a sharp increase in average assets growth by nearly 10% pre-violation for all violators and frequent violators. At the quarter of the debt covenant violation asset growth reverses and drops for all violators by an estimated 4.1% (cf. column (3), Figure 3) and for frequent violators by an estimated $|-3.6\% - 1.6\%| = 5.2\%$, at the 1% significance level, within one year post-violation. Column (4) in Table 3 also shows that frequent violators (one-time violators) have at times of no debt covenant violation, on average and c.p., 1.6% lower (1.7% higher) growth rate of total assets compared to non-violator firms, with the difference statistically significant at the 1% (5%) level. For one-time violators the pattern in Panel A, Figure 2 differs significantly. Namely, while median total assets fall by ($e^{5.61-5.56} \approx$) \$1.05 million between $q = -1$ and $q = 0$, the average asset growth appears to stagnate. There is no discernible change in the positive average growth rate of total assets in the four quarters post-violation compared to pre-violation. The estimated effect weakens this thesis somewhat, but still supports our Hypothesis II: for one-time violators a new debt covenant violation is, c.p., associated with an average 1.9% reduction in asset growth, compared to an average 4.1% reduction (cf. column (3), Figure 3) when not dis-aggregating all violator firms by group.

Panel B of Figure 2 shows that the pattern of PPENT growth for all violators and the two violator groups are similar to Panel A. In combination with the results of column (8) of Table 3 we find that a new debt covenant is associated with statistically significant estimated decline of 0.7% in the growth rate of PPENT for one-time violators. At times of no debt covenant violation, one-time violators have on average a 2.2% higher growth rate in PPENT than non-violators. Frequent violators do not exhibit a statistically significant lower average growth rate in PPENT than non-violators.

As can be inferred from Panel C of Figure 2, the time paths of the median and mean of the CAPEX-to-total assets ratio around a new debt covenant violation for all violator and frequent violator firms are almost identical. Column (12) of Table 3 shows that the estimated effect for both groups is, in fact, of the same magnitude. Capital expenditures, already decreasing pre-violation, drop by, c.p., an estimated 0.6 percentage points in the year after a new debt covenant was reported. For one-time violators this negative effect is leveled off by an estimated statistically significant 0.2 percentage points post-violation, c.p., largely due to the reversal of the trend two quarters after the new debt covenant violation was reported.

Panel D of Figure 2 shows that the mean R&D-to-assets ratio aggressively rises pre-violation and falls sharply post-violation, the median value, however, shows no discernible trend. The estimated impact presented in column (16) of Figure 3 indicates that the violation of new debt covenants has no statistically significant impact on the ratio of R&D to total assets for either violator group. Ultimately, these findings support Hypothesis II partially: new debt covenant violations exert statistically significantly less influence on firm's investment policy if the firm classifies as one-time violator. The R&D-to-assets ratio remains unaffected by a new debt covenant violation. Frequent violators experience statistically significant a lower growth rate in total assets after a new debt covenant violators.

5 Conclusion

Using two variants of panel data analysis, we strengthen the extant evidence of Nini et al. (2012) and Chava and Roberts (2008) that debt covenant violations trigger changes in firm's investment policy. We find that covenant violations lead to a reduction in the growth rate of total assets and net property, plant & equipment and a decrease in capital expenditures to asset ratio within one year post-violation. The results suggest that non-financial U.S. firms engage in asset sales, divestitures and investment cutting in response to a debt covenant violation. Though they do not appear to lower their relative R&D expenditures one year after the violation as a result of strengthened restrictions imposed by creditors in amended loan agreements. We also demonstrate that the level of creditor influence varies. While the effect of debt covenants on firms' financial policies is more larger in size for firms that have reported more than one covenant violation, it is smaller for one-time violator firms. Thus, our results show how a one-time violation mitigates the creditor's influence and demonstrates the resulting changes in the firm's investment policy.

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Appendix: List of Abbreviations

Capitalized names denote variables obtained from Compustat database. Italicized names denote the names of the variables as they appear on S&P Capital IQ's database, unless another source is indicated.

Abbreviation	Definition [Computation]
No.	Number
obs.	Observations
Adj.	Adjusted
OLS	Ordinary Least Squares
SEC	Securities and Exchange Commission
Financial covenant violation	A dummy variable that equals 1 if the given firm has violated a financial covenant and discloses this in its 10-K or 10-Q SEC filings or if it has been granted a reported violation waiver for the quarter in question, zero otherwise. Variable *viol* obtained from supplementary data of Nini et al. (10), available at `https://amirsufi.net/chronology.html` (File name and direct download: cstatviolations_nss_20090701.dta). For more information on the extraction of the variable *viol*, please refer to the separate, supplementary data file in the appendix of (10), which is available in the online version of the article on https://academic.oup.com. The term financial covenant (violation) and debt covenant (violation) are used interchangeably in this paper.
New financial covenant violation	A dummy variable that equals 1 if the given firm reports a financial covenant violation (as previously defined) in the given quarter (t), but was continuously observed to be in no financial covenant violation for at least four previous consecutive quarters (t-1, t-2, t-3, t-4), and zero otherwise. The term new financial covenant (violation) and new debt covenant (violation) are used interchangeably in this paper.
One-time violator	A dummy variable that equals 1 if the given firm has reported exactly one financial covenant violation (as previously defined) in any quarter over the full sample period applied, 1997-2008 starting and ending with the second calendar quarter, and zero otherwise.
Frequent violator	A dummy variable that equals 1 if the given firm has reported more than one financial covenant violation (as previously defined) in any quarter over the full sample period applied, 1997-2008 starting and ending with the second calendar quarter, and zero otherwise.
Assets	*Total Assets* in each quarter in $ millions.

(Continued)

Average assets	The sum of total assets and lagged total assets, divided by two, in each quarter in \$ million [(*Total Assets* + lagged *Total Assets*)/2].
First Differences Ln(Assets)	Difference between logarithmized total assets and logarithmized total assets with a lag of one quarter, in each quarter in \$ millions [log(*Total Assets*) − log(lagged *Total Assets*)].
Assets (4 leads)/ Assets	Ratio of total assets with a lead of four quarters to total assets, in each quarter [$(Total\ Assets_{t+4})/(Total\ Assets_t)$].
Change in Ln(Assets)	Difference between logarithmized total assets with a lead of four quarters and logarithmized total assets, in each quarter in \$ millions [$\log(Total\ Assets_{t+4}) - \log(Total\ Assets_t)$].
PPENT	*Net Property, Plant & Equipment* in each quarter in \$ millions.
Change in Ln(PPENT)	Difference between logarithmized *Net Property, Plant & Equipment* with a lead of four quarters and logarithmized *Net Property, Plant & Equipment*, in each quarter in \$ millions.
PPENT (4 leads)/ PPENT	Ratio of *Net Property, Plant & Equipment* with a lead of four quarters to *Net Property, Plant & Equipment*, in each quarter.
PPENT/ Assets	Ratio of *Net Property, Plant & Equipment* and *Total Assets*, in each quarter, given ratio is between (−1) and 1.
First Differences (PPENT/ Assets)	Difference between *Net Property, Plant & Equipment* to *Total Assets* and its one-quarter-lagged ratio, in each quarter in \$ millions.
Operating cash flow	Annualized operating income before depreciation and amortization, in each quarter in \$ millions [(*Operating Income*+*Depreciation & Amortization*) * 4].
Operating cash flow/ Average assets	Ratio of operating cash flow (as previously defined) and average assets (as previously defined), in each quarter, under the constraint that the (lagged) sum of *Operating Income* and *Depreciation & Amortization* divided by (lagged) *Total Assets* takes a value in between −1 and 1, in each quarter.
Leverage	Ratio of total debt (long-term debt plus debt in current liabilities) to total assets, in each quarter, given ratio is between 0 and 1 [*Long Term Debt* + DLCQ)/*Total Assets*].
Interest expense	Annualized interest expense multiplied by (−1), in each quarter in \$ millions. [*Interest Expense* * (−4)].
Interest expense/ Average assets	Ratio of interest expense (as previously defined) and average assets (as previously defined), in each quarter, under the constraint that (lagged) *Interest Expense* * (−1) divided by (lagged) *Total Assets* take a value in between −1 and 1, in each quarter.
Net worth/ Assets	Ratio of *Total Parent Equity* to *Total Assets*, in each quarter, given ratio is between (−1) and 1.

(Continued)

Current ratio	*Current Ratio* in each quarter.
Total sales	*Revenue* in each quarter in $ millions.
Market-to-book	Ratio of market value of assets (market value of equity minus book value of equity plus total assets) to total assets in each quarter [(PRCCQ $*$ *Common Shares Outstanding* $-$ (*Total Assets* $-$ *Total Liabilities* $+$ TXDITCQ) $+$ *Total Assets*)/*Total Assets*].
CAPEX	Annualized capital expenditures multiplied by (-1), in each quarter in $ millions [*Capital Expenditures* * (-4)].
CAPEX/ Average Assets	Ratio of CAPEX (as previously defined) and average assets (as previously defined), in each quarter, under the constraint that (lagged) *Capital Expenditures* * (-1) divided by (lagged) *Total Assets* take a value in between -1 and 1, in each quarter.
R&D/ Assets	Ratio of research and development expenditures to total assets in each quarter, given ratio is between 0 and 1 [(*R&D Expense*/*Total Assets*].